

Nikitha Kasipati

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EDUCATION

Rochester Institute Of Technology

Rochester, NY

Masters of Science in Data Science (minor in Business Analytics) - GPA 3.95

Sep 2023 - May 2025

Relevant coursework: Advanced Machine Learning, Database Management Systems, Applied Statistics, Optimization and computational linear algebra, Software engineering and development, Big data, Marketing analytics, Data warehousing

Vellore Institute Of Technology

Vellore, India

Bachelors of Technology in Computer Science and Engineering - GPA : 4.0

Relevant coursework: Strong foundation in programming languages, Data structures and algorithms, Statistical techniques for business analysis, Intro to Artificial Intelligence, Natural language processing, Data visualization

TECHNICAL SKILLS

Languages: Python, SQL, R, Scala, Java, C++, HTML, CSS

Libraries: NumPy, Pandas, Scikit-Learn, SciPy, Matplotlib, Seaborn, TensorFlow, PyTorch, Keras, NLTK, LangChain

Tools: JMP Pro, Power BI, Tableau, Git, Microsoft Office (Word, Excel, PowerPoint)

Big Data & Cloud Platforms: Apache Spark, Databricks, BigQuery, Snowflake, AWS, GCP

Databases: MySQL, MongoDB, NoSQL, PostgreSQL, Azure SQL, Redis

Methods/Techniques: Deep Learning, Data mining, Fine-tuning LLMs, Data Modeling, Predictive Modeling, Data Visualization, Time Series Forecasting, Data Acquisition, Data Warehousing, Data Wrangling, Data Engineering

WORK EXPERIENCE

Arhasi Inc

Sep 2024 - Present

Data Scientist, Frisco, TX

- **Developed multi-agent RAG frameworks and an AI/BI dashboard** using FastAPI, LangChain, and Google Vertex AI, implementing dynamic **prompt engineering for SQL generation and data visualization** to optimize insights across geospatial, healthcare, and FinTech applications.
- **Integrated BigQuery, MongoDB, and Redis** with CI/CD pipelines to automate data retrieval, schema analysis, metadata generation, and caching, significantly enhancing application responsiveness and scalability.
- Created a RAG-based **Data Lineage Agent** to automatically detect data hierarchies and dependencies, leveraging **Neo4j** to visualize relationships and enhance understanding of data flow across complex pipelines.
- Utilized **Optical Character Recognition (OCR)** technology to convert diverse document formats into editable and searchable data, improving data accessibility and operational efficiency.

Maxime Tools

Jun 2024 - Aug 2024

ML Engineer Intern, San Francisco, CA

- **Engineered and optimized RAG models** in collaboration with the ML team, leveraging ``autorag`` and ``llama-index`` libraries for high-performance retrieval-augmented generation.
- **Enhanced retrieval precision** by implementing and fine-tuning retrieval pipelines with ``tqdm``, achieving a 15% boost in content accuracy and relevance.
- **Executed rigorous model evaluation** via custom Python scripts, conducting iterative experimentation cycles that improved model recall by 10% through targeted adjustments in retrieval and generation components.

Infinite Uptime

Jan 2024 - May 2024

Graduate Student Researcher, Berkeley, CA (Remote)

- Implemented **time series forecasting and anomaly detection algorithms** to predict equipment failures, reducing downtime by 20%.
- Conducted **data preprocessing, feature engineering, and deployed models** into production environments, enhancing operational efficiency and providing actionable insights through detailed reports and visualizations.

New York University

May 2023 - Aug 2023

Summer Research Intern, New York City, NY

Fine tuning GPT3: Enhancing SQL Auto-grading and Feedback Capabilities

- Developed and implemented a comprehensive strategy to improve the SQL auto-grading and feedback capabilities of GPT3, resulting in a 30% reduction in grading time for instructors.
- Utilized **OpenAI's CLIs interface** to **fine-tune Ada and Davinci models**, resulting in an improvement in model accuracy compared to previous versions.
- Leveraged **LLMs** and **transformers** from the **HuggingFace library**, incorporating **LoRA** to enhance model performance

Scale Global

Jul 2021 - Jun 2022

Data Analyst, Hyderabad, India

- Spearheaded inventory optimization through the implementation of **time series forecasting techniques** using ARIMA and LSTM, resulting in a 12% reduction in excess stock.
- Worked on building a **Tableau dashboard** of sales reports, to give better insights by geography, product grouping, category, by principal and analyze trends accordingly.
- Used machine learning models to pinpoint and bridge critical e-commerce acceleration gaps, resulting in a 15% increase in sales.
- Employed **A/B testing** and **Bayesian statistical analysis** to refine marketing strategies, leading to improvement in customer acquisition and retention rates.

World Wide Commercial Ventures Limited (WWCVL)

Jan 2021 - Apr 2021

IT Intern, Lagos Nigeria

- Optimized **SQL** database queries for a high-traffic application, achieving a 30% reduction in execution time and enhancing server throughput by 20%.
- Developed innovative **Power BI** dashboards for **data visualization**, significantly enhancing decision-making efficiency and slashing report generation time by 40%.
- Developed **ETL (extract, transform, load)** data pipelines with **Databricks, Apache Spark, and Apache Kafka**, boosting data accuracy, streamlining efficiency, and reducing manual effort by over 50%.
- Initiated and executed a **Unique Products Tracker with Python, Pandas, and PostgreSQL**, incorporating **Angular JS** for user authentication, and crafted a **jQuery script** for efficient data filtering, resulting in streamlined product tracking and data management.

ACADEMIC PROJECTS

• Popularity based Music recommendation system

- Cleaned and structured interaction count data, integrating detailed user and song metadata to **create robust user-item interaction matrices**, improving recommendation accuracy by 15%.
- Developed and implemented a music recommender system using **Apache Spark's ALS algorithm and the LensKit toolkit**, resulting in a 20% increase in user engagement and satisfaction, with an accuracy rate of 90%.